Ali Shafiee

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alishaffiee.github.io | Codeforces | LinkedIn | IOI Profile | Google Scholar

IOI silver medalist and ICPC World Finalist. PhD researcher in stochastic control, robust optimization, and algorithmic game theory.

Education

Institute of Science and Technology Austria (ISTA) — Vienna, Austria
PhD in Computer Science, grade average 1.0 (Austrian scale, 1.0 highest) — *Sep. 2025 – 2028 (expected)*

Sharif University of Technology — Tehran, Iran
BSc in Computer Engineering, GPA 3.68/4 (17.03/20) — *Sep. 2018 – Aug. 2023*

Publications

Author lists are alphabetical, per convention in theoretical computer science.

- **Randomise Alone, Reach as a Team**. L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *CAV 2026*. [paper]
- **Qualitative Analysis of ω-Regular Objectives on Robust MDPs**. A. Asadi, K. Chatterjee, E. Goharshady, M. Karrabi, **A. Shafiee**. *AAAI 2026*. [paper]
- **Limit-Sure Reachability for Small-Memory Policies in POMDPs is NP-Complete**. A. Asadi, K. Chatterjee, R. Saona, **A. Shafiee**. *UAI 2025*. [paper]
- **Quantitative Analysis of ω-Regular Robust MDPs**. A. Asadi, K. Chatterjee, E. K. Goharshady, M. Karrabi, A. Montaseri, **A. Shafiee**. *AAAI 2027, under review*. [paper]
- **On the Complexity of Discounted Robust MDPs with L_p Uncertainty Sets**. A. Asadi, K. Chatterjee, A. Montaseri, **A. Shafiee**. *NeurIPS 2026, spotlight*. [paper]
- **Limit-sure Winning in Team Games without Shared Randomness**. L. Brice, T. A. Henzinger, A. Montaseri, **A. Shafiee**, K. S. Thejaswini. *AAMAS 2027, under review*. [paper]
- **Mechanism Design for Multi-Resource Nakamoto Consensus**. N. Babashah, M. A. Baig, K. Pietrzak, **A. Shafiee**. *WINE 2026, under review*.

Competitive Programming

- **Codeforces**: **top 50 worldwide in 2019**; International Grandmaster, peak rating 2800 (handle `ToTLeS`)
- **Silver Medal, International Olympiad in Informatics (IOI)**, 2017: 76th of 300+ contestants from 80 countries
- **33rd Place, ACM-ICPC World Finals**, Moscow 2021: among 10,000+ teams from 110 countries
- **Gold Medal, ACM-ICPC West Asia Regional Contest**: 1st place 2019, 2nd 2018, 3rd 2020; 100+ teams each year
- **Silver Medal, Asia-Pacific Informatics Olympiad (APIO)**, 2018 and 2017: 20+ participating countries
- **Gold Medal, Iranian National Olympiad in Informatics**, 2016: top 8 of 10,000+ students nationally
- **Bronze Medal, Iranian Mathematical Olympiad**, 2017: top 80 of 20,000+ students nationally

Research Experience

Institute of Science and Technology Austria (ISTA) — Vienna, Austria
Research Intern and Online Collaborator, then PhD Researcher, advisor Prof. Krishnendu Chatterjee — *Apr. 2024 – present*

- Proved NP-completeness of limit-sure reachability for small-memory POMDP policies, resolving an open complexity question; published at UAI 2025.
- Derived algorithms and complexity bounds for robust MDPs under ω-regular and discounted objectives, leading to four further papers.
- Implemented solvers in C++ and Python and benchmarked them against existing approaches on random and structured MDP families, scaling to instances with thousands of states.

Industry Experience

Quera, Iran's largest competitive-programming and technical-education platform — Tehran, Iran
Contest Manager, Content Manager, Director of CS Education, Business Line Manager — *Mar. 2019 – Aug. 2025*

- Owned a business line end to end (Oct. 2024 – Aug. 2025); previously led a 40-person division spanning six technical teams (back-end, front-end, data, blockchain, algorithms, fundamentals) and set the education product roadmap.
- Designed and shipped 12+ bootcamps and courses in C++, Python, SQL, data analysis and machine learning, and ran 20+ large-scale programming contests including sponsored hiring hackathons.

Teaching & Service

- **Scientific Committee**, ACM-ICPC West Asia Regional Contest (2024 – present); **Host Scientific Committee**, IOI 2019, Baku
- **Problem-setter** (2017 – present) for the Iranian National Olympiad in Informatics, Codecup and Quera; **olympiad coach** (2017 – 2023), 50+ students trained for INOI and the IOI
- **Teaching Assistant**, Sharif University: Design of Algorithms, Data Structures & Algorithms, Fundamentals of Programming

Skills

Programming: C++, Python, Rust · **Numerical & ML**: NumPy, SciPy, pandas, scikit-learn, PyTorch, JAX
Mathematics: probability, stochastic processes, Markov decision processes, optimization, game theory
Tools: Git, Linux/Bash, SQL, Docker, LaTeX · **Languages**: Persian (native), English (TOEFL 110/120)